Size and Environment Dependence of Plasmon Resonance in Gold Nanostructures

COMSOL Multiphysics Study

Objective

To quantitatively investigate the dependence of plasmon resonance on **metasurface element size** and **dielectric environment** in a **periodic gold nanostructure array (metasurface)** modeled using **periodic boundary conditions**, capturing inter-element coupling effects.

Methodology

- COMSOL Multiphysics (Wave Optics Module)
- Structure: Periodic array of gold nanostructures (metasurface configuration)
- Boundary conditions: Periodic boundary conditions applied in the lateral directions
- Nanostructure - Au taken from Material Library: $D = 100, 150, 200$ nm, $h = 10$ nm
- Excitation: linearly polarized plane wave (normal incidence)
- Environments: free space ($n = 1.0$), SiO_2 substrate, water ($n = 1.33$)
- Boundary: Perfectly Matched Layers (PML)
- Output: **transmittance vs wavelength**
  The use of periodic boundary conditions models an infinite array, introducing inter-particle coupling effects that can modify the resonance compared to isolated nanoparticles.

Results and Discussion

1) Size Dependence: As shown in Fig. 1, increasing element size produces a **monotonic redshift** in plasmon resonance.

Resonance is identified from minima in transmittance, corresponding to increased absorption and scattering losses.

The resonance dip shifts from 748 nm to 973 nm ($\Delta = 225$ nm). Also, the full width at half maximum (FWHM) of the resonance increases with element size, indicating enhanced radiative damping and reduced spectral selectivity.

Interpretation: The redshift arises from increased effective oscillation length.

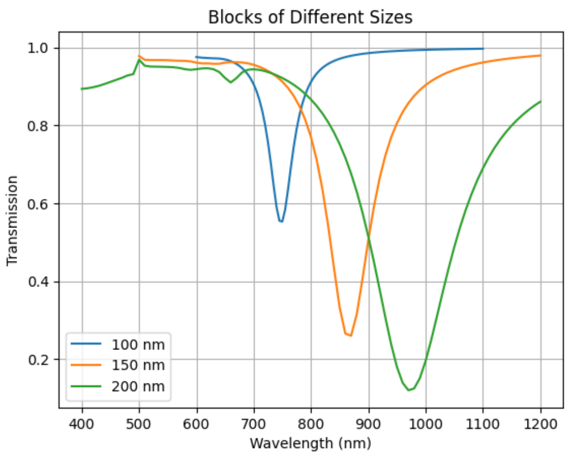

Figure 1: Transmittance spectra for metasurface element diameters (100–200 nm), showing redshift of resonance dip and spectral broadening with increasing size.

2) Environmental Effect: Figure 2 shows the effect of surrounding refractive index for a fixed element size ($D = 150$ nm). The resonance dip shifts from 866 nm to 1108 nm ($\Delta = 242$ nm). The substrate case exhibits an intermediate shift, reflecting partial dielectric screening at the interface.

Notably, the environmental shift (~242 nm) is slightly larger than the size-induced shift (~225 nm), indicating a strong environmental influence on resonance. In the metasurface configuration, the substrate and surrounding medium collectively influence the effective refractive index experienced by the array.

Interpretation: Increasing refractive index shifts the resonance to longer wavelengths, as predicted by classical plasmonic theory. This behavior is consistent with the plasmon resonance condition $\Re[\varepsilon_m] = -2\varepsilon_d$.

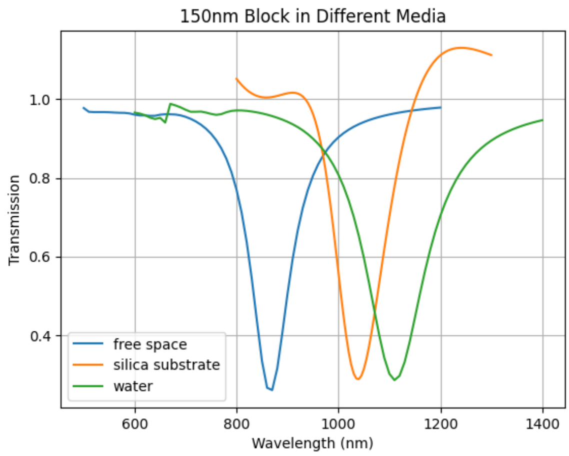

Figure 2: Transmittance spectra for different environments (air, substrate, water), showing refractive index dependence of resonance dip.

3) Geometric Dependence: Variations in metasurface element geometry produce measurable differences in spectral response.

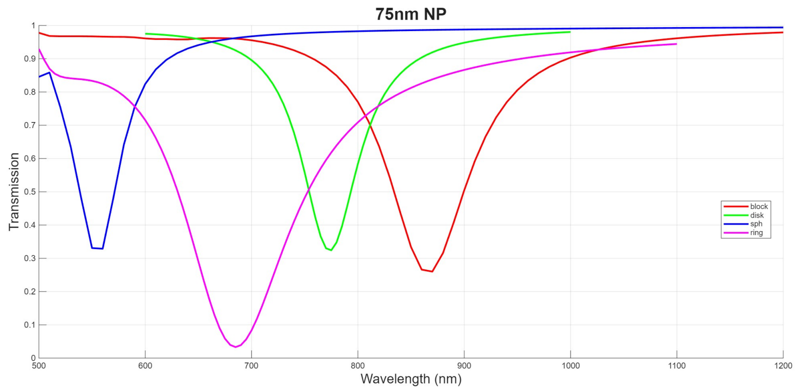

Figure 3: Transmittance spectra for different element geometries (block, sphere, disk, ring; where D = 75 nm), showing geometry-dependent variation in resonance position and spectral profile.

The transmission minima occur at approximately 550 nm (sphere), 690 nm (ring), 770 nm (disk), and 860 nm (block). Block-like structures exhibit larger redshifts due to edge-induced field localization.

The observed trends are consistent with classical plasmonic theory.

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Key Takeaway

Plasmon resonance in gold nanostructures shows clear dependence on size, environment, and geometry, consistent with classical theory.

Tools

COMSOL Multiphysics — Electromagnetic Simulation — Plasmonics